Real-Time Hand Gesture Recognition Using EMG Signals

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Introduction and Objectives

Surface electromyography (sEMG) signals provide an intuitive way to recognize hand gestures without the need for cameras or vision-based systems. This project aims to develop a real-time gesture recognition system using EMG signals captured by the Myo armband. Real-time recognition is challenging due to the noisy nature of EMG data, variability between users, and the need for low-latency predictions. To address these challenges, we designed a lightweight machine learning pipeline combining careful preprocessing, feature extraction, and robust model selection

Methodology

We developed a real-time gesture recognition pipeline consisting of data collection, signal preprocessing, feature extraction, class balancing, and model training.

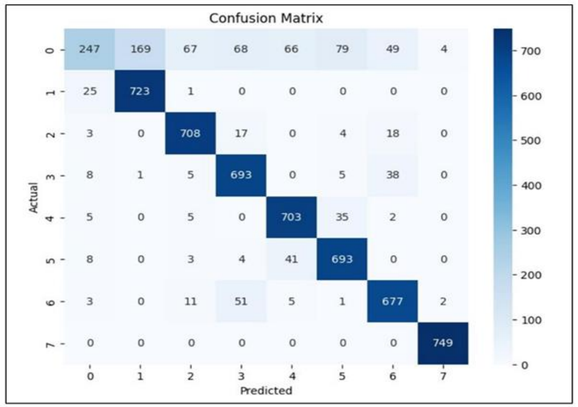
1. Data Collection: EMG signals were recorded using the Myo armband (8 channels) across multiple subjects.
2. Signal Preprocessing: Signals were normalized (zero-mean, unit-variance) to handle variability. Gesture segments were detected using a spectrogram-based energy method.
3. Feature Extraction: Simple time-domain features were extracted for each segment: Mean Value, Root Mean Square (RMS),Variance. Features from all channels were concatenated into a single vector.
4. Class Balancing and Model Training: Stratified sampling and class weighting (class_weight='balanced') addressed class imbalance. Three models were trained and compared: Random Forest (100 trees) / Support Vector Machine (RBF kernel, C=1.0)/ Decision Tree (baseline).

The lightweight pipeline ensures fast processing and low-latency gesture recognition suitable for real-time applications.

Results and Discussion

The Random Forest classifier

- achieved the highest test accuracy, reaching 86.6%, and a weighted F1-score of approximately 0.85 across eight gesture classes.
- Its ensemble structure provided robust generalization, reducing the impact of noise and inter-subject variability, and maintaining real-time processing speeds with minimal latency.

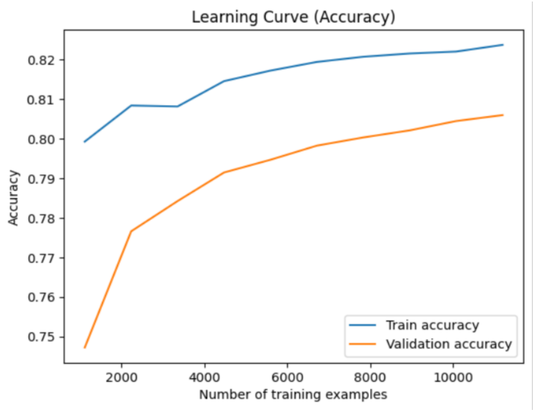


The matrix highlights strong recognition for most gestures, particularly class 7 (near 100% accuracy). However, confusion occurred between class 0 and class 1, reducing the recall for class 0 to approximately 33%, likely due to overlapping muscle activation patterns.

- The learning curve of the Random Forest model showed 100% training accuracy, but validation accuracy plateaued around 85–87%, indicating moderate overfitting despite the ensemble effect.
- Nevertheless, the Random Forest model demonstrated the best trade-off between classification performance and low-latency real-time inference, making it the most suitable choice for deployment in our real-time EMG-based gesture recognition system.

The Support Vector Machine model

- achieved a test accuracy of 81.8%, with a weighted F1-score of approximately 0.78.
- SVM exhibited better generalization compared to Random Forest, with closer training and validation accuracies throughout training, and lower risk of overfitting.



The curve demonstrates that SVM maintained consistent learning, with a small, stable gap between training and validation accuracies. Validation accuracy steadily improved as more data was introduced, reaching around 80%. This suggests that the model was slightly underfitting the problem and could benefit from a higher model capacity (e.g., tuning hyperparameters or extracting richer features). However, SVM still struggled with subtle gesture distinctions, particularly between class 0 and class.

Despite its slightly lower peak accuracy compared to Random Forest, the SVM model is highly efficient, computationally lightweight, and well-suited for real-time inference, making it a strong candidate for real-time gesture recognition systems when robustness and low latency are priorities.

The Decision Tree classifier

- achieved a test accuracy of 77.9% and a weighted F1-score of about 0.78.
- It exhibited severe overfitting, reaching 100% training accuracy but poor validation performance.
- Misclassifications were widespread across multiple gesture classes, not only between class 0 and class 1.
- Although it was the fastest model computationally, its low reliability made it unsuitable for real-time gesture recognition.

All results and analyses presented above correspond to training and evaluation with class 0 included. A second experiment excluding class 0 led to overall higher model accuracies, but for consistency and completeness, this poster focuses on the case where the full set of gesture classes was used.

Conclusion

We developed a real-time gesture recognition pipeline using surface EMG signals collected with the Myo armband. By combining careful preprocessing, time-domain feature extraction, and lightweight machine learning models, we achieved accurate and low-latency classification across multiple users. The Random Forest model provided the best results, reaching 86.6% test accuracy, while the SVM offered good generalization with slightly lower accuracy. The Decision Tree model, although fast, suffered from severe overfitting and poor generalization. Challenges such as confusion between similar gestures (e.g., class 0 vs. class 1) impacted performance, but overall, the system maintained robust real-time capabilities. Future improvements include exploring richer features (e.g., wavelets, frequency-domain analysis), adaptive learning per user, and combining EMG with inertial data to further enhance recognition accuracy.